

Success Story

Big Data Solution on AWS



Shree Partners collaborated with one of the world's largest travel management consortiums to process large amounts of data using big data technologies on AWS cloud

The Client

The client is the world's leading business travel provider, offering end-to-end corporate travel management, meeting and booking solutions.

Requirements and Challenges

Because the client had numerous data sources, it needed to consolidate and leverage the data to glean important insights and make smarter business decisions. Key objectives were to make the data available more quickly, apply new or changed business rule validation/transformation to incoming data, and improve data cleansing in the overall workflow. The client was looking for an integrated, high-quality tool to replace legacy systems to quality-check data with different formats and sources. The client also wanted to create a unified data system with enhanced security. Rules execution was needed both for real-time user updates and batch processing

Shree Partners Solution

To enable better data integration – and ensure a more scalable system, Shree built a cost-effective data lake using big data technologies with Apache Spark, Hive, ORC, Cassandra, Sqoop, Java, Python, Scala and Drools (BRMS).

- Data ingestion and modeling: Imported files from legacy systems into data lake using File Mover and a custom Jenkins Job; created models for GRO 3.x format; mapped legacy system formats to GRO3.x model
- Created, configured and implemented rules engine; upgraded to Drools (Kie Workbench); created and configured Drools Projects and Repository; rules translation matrix included field mapping for legacy system; implemented security on Drools Project and menus
- Data transformation and quality checks: Created custom project to take the data into Spark; applied rules to data in memory
- DV models: Exposed data lake to external clients through DV; implemented security on DV layer

Business Benefits

- Single global application solution for data quality
- Harnessed the power of parallel processing
- Data ingestion from different data sources (files, DB2, SQL, rest APIs)
- PCI compliant with Row-level Security
- Time to make data live was reduced from days to within one hour
- Users can create rules and integrate seamlessly with workflow (no IT development required)
- Mapping, tokenizing, validating and transforming data within an hour (potential for additional decreases with more nodes in the cluster)
- Functionality of data can be changed by the user (through UI) during quality check process
- Future capability to reduce manual/scripted data changes to zero by using ML on logs of changes

About Shree Partners

Shree Partners is a digital transformation and technology solutions company that helps customers increase profitability by designing, developing, implementing and maintaining innovative business solutions that drive revenue and reduce cost.

Founded in 2008, our mission is to deliver high-performance applications by leveraging deep domain competencies. We serve diverse client businesses in travel and hospitality, healthcare, retail and financial services.

Our innovative, technology-based solutions feature a combination of software design, development, implementation and global support services. Our expertise in disruptive technologies – cloud, mobility, analytics and infrastructure – have helped us consistently deliver solutions that add significant business value to clients.